

COTTBUS, Germany

WMO: 104960

Lat: 51.7761N

Lon: 14.3169E

Elev: 70

StdP: 100.49

Time Zone: 1.00 (EUC)

Period: 90-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-9.0	-14.4	1.1	-11.5	-11.4	1.4	-8.5	10.8	8.0	9.8	7.6	2.4	70	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.6	31.1	19.4	29.2	18.5	27.5	17.9	20.7	29.2	19.7	26.8	18.7	25.3	3.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.0	22.7	17.1	12.3	22.0	16.2	11.6	21.6	59.7	29.5	56.3	26.8	53.3	25.4	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.5	6.5	-16.3	35.9	2.7	1.9	-18.3	37.2	-19.9	38.3	-21.4	39.4	-23.3	40.7		
(5)			-16.1	22.5	2.9	0.5	-18.1	22.8	-19.8	23.1	-21.4	23.4	-23.5	23.8	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	0.7	2.7	5.1	9.3	14.1	17.4	19.4	19.8	14.4	10.0	4.4	0.7	(6)
(7)		DBStd	7.92	4.89	5.19	3.64	4.41	3.99	3.60	3.05	3.22	2.98	4.21	3.69	4.65	(7)
(8)		HDD10.0	1243	290	208	157	64	8	0	0	2	54	172	288		(8)
(9)		HDD18.3	3262	548	438	409	273	140	61	23	22	122	260	419	546	(9)
(10)		CDD10.0	1196	1	3	6	43	137	221	292	304	134	53	2	1	(10)
(11)		CDD18.3	173	0	0	0	2	10	32	57	68	4	0	0	0	(11)
(12)		CDH23.3	1854	0	0	0	34	164	349	569	679	56	2	0	0	(12)
(13)		CDH26.7	577	0	0	0	4	38	113	185	230	7	0	0	0	(13)
(14)	Wind	WSAvg	2.8	3.1	3.6	3.3	2.7	2.5	2.5	2.4	2.2	2.5	2.8	2.9	2.9	(14)
(15)	Precipitation	PrecAvg	597	45	37	41	33	56	57	81	64	49	39	48	47	(15)
(16)		PrecMax	893	76	78	97	88	121	126	192	224	122	88	138	99	(16)
(17)		PrecMin	393	3	6	15	2	10	9	7	11	11	8	2	15	(17)
(18)		PrecStd	109	18	18	23	19	30	28	46	43	27	21	27	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	15.9	19.1	27.2	30.0	33.2	34.4	33.9	27.5	23.2	14.6	12.5	(19)
(20)			MCWB	9.1	10.4	11.8	15.5	17.4	20.8	21.1	20.9	17.5	16.7	11.1	9.9	(20)
(21)		2%	DB	10.4	13.5	15.9	23.3	27.6	29.5	30.9	31.0	24.8	20.6	12.6	10.3	(21)
(22)			MCWB	7.9	9.2	10.2	14.5	17.0	18.6	19.5	19.5	16.3	15.2	9.6	8.1	(22)
(23)		5%	DB	8.7	11.6	13.0	19.8	24.9	27.3	28.6	29.1	22.6	18.1	11.2	8.7	(23)
(24)			MCWB	6.4	7.9	9.0	12.7	15.5	17.8	18.5	18.6	16.3	13.9	8.9	6.7	(24)
(25)		10%	DB	7.1	9.9	11.0	17.1	22.6	25.0	26.4	27.3	20.2	16.1	9.6	6.8	(25)
(26)			MCWB	5.0	6.9	7.6	11.1	14.6	16.8	17.7	18.2	15.2	12.6	7.7	5.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	11.0	12.3	16.3	18.8	21.7	21.9	21.6	19.0	17.2	11.9	9.7	(27)
(28)			MCDWB	12.2	15.2	17.5	25.2	26.8	30.3	31.7	29.8	24.6	22.1	14.8	11.4	(28)
(29)		2%	WB	8.1	9.7	10.6	14.9	17.6	19.9	20.5	20.7	17.9	15.6	10.0	8.3	(29)
(30)			MCDWB	10.2	13.2	14.7	23.4	25.6	27.2	28.9	29.2	23.0	19.3	12.0	10.2	(30)
(31)		5%	WB	6.5	8.1	9.3	13.1	16.5	18.6	19.4	19.7	16.8	14.3	8.9	6.8	(31)
(32)			MCDWB	8.5	10.9	12.7	18.8	23.2	25.5	26.2	27.1	20.9	17.3	10.8	8.5	(32)
(33)		10%	WB	5.1	7.0	8.0	11.6	15.1	17.5	18.4	18.8	15.6	13.1	7.9	5.3	(33)
(34)			MCDWB	6.8	9.6	10.4	16.3	21.0	23.6	24.9	25.3	19.7	15.8	9.6	6.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.4	7.0	8.0	10.4	11.5	10.7	10.8	11.6	9.5	8.2	5.6	4.9	(35)
(36)			MCDBR	6.9	9.1	11.9	14.4	16.1	15.1	15.1	15.0	13.5	11.2	6.8	6.6	(36)
(37)			MCWBR	5.3	6.1	7.5	7.3	7.7	6.5	6.3	6.2	7.1	6.8	5.0	5.3	(37)
(38)		5% WB	MCDBR	6.8	7.8	10.8	12.9	14.0	13.0	12.9	13.0	11.0	9.9	6.7	6.6	(38)
(39)			MCWBR	5.4	5.8	7.2	6.7	7.0	6.3	5.7	5.8	6.3	6.5	5.2	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.322	0.347	0.389	0.421	0.420	0.416	0.435	0.429	0.389	0.372	0.347	0.315	(40)	
(41)		taud	2.335	2.314	2.217	2.193	2.243	2.283	2.236	2.272	2.354	2.365	2.370	2.352	(41)	
(42)		Ebn at Noon	674	762	793	813	834	840	814	798	790	723	637	618	(42)	
(43)		Edh at Noon	64	85	113	130	130	126	130	120	98	81	62	54	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.71	1.40	2.44	4.04	4.92	5.22	4.95	4.32	3.11	1.77	0.82	0.52	(44)	
(45)		RadStd	0.10	0.26	0.32	0.56	0.58	0.41	0.61	0.41	0.41	0.23	0.13	0.06	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	+0.38	N/A	N/A	+57

Nomenclature: See separate page